

lifetime of at least 15 years. Its solar arrays and six nickel-cadmium batteries can provide an average of 3 kilowatts of electrical power. Its side docking ports accommodate Russian Soyuz piloted spacecraft and unpiloted Progress resupply spacecraft. The module's 16 fuel tanks combined can hold more than 6 tons of propellant. The attitude control system for the module includes 24 large steering jets and 12 small steering jets. Two large engines were available for reboosting the spacecraft and making major orbital changes.

Unity

The Unity node, the first major US-built component, made the station truly 'international' in December 1988. It is a connecting passageway to living and work areas of the International Space Station, including the US Laboratory Module and an airlock. It has six hatches that serve as docking ports for the other modules. Fabricated out of aluminium, it is 5.5 metres long, 4.6 metres and contains more than 50,000 mechanical items, 216 lines to carry fluids and gases, and 121 internal and external electrical cables using 9.7 kilometres of wire.

Zvezda

In July of 2000, Russia launched its second module which would be the first habitable headquarters of the ISS. Meaning "star" in Russian, Zvezda provided the early station living quarters (as well as the functional centre of the current Russian portion of the ISS), life support system, electrical power distribution, data processing system, flight control system and propulsion system. It also provided a communications system that includes remote command capabilities from ground flight controllers, although many of these systems are supplemented or replaced by other components now.

Zvezda made it possible for cosmonauts and astronauts to live aboard the space station and in October 2000, Bill Shepherd, Yuri Gidzenko and Sergei Krikalev launched into history as the first crew to live on the ISS. From the moment the hatch of their Soyuz spacecraft opened and they entered the fledgling space station, there have been people living and working in orbit, 24 hours a day, seven days a week, 365 days a year.

Integrated Truss Structure

Colloquially called the "backbone" of the ISS, the Integrated Truss Structure is a complex alignment of various parts that allow for mounting of solar arrays, carriers, etc. Considering that everything on the ISS runs on electricity, generating power became the most important part of the early stages of the programme.